

Designing an AI-Powered Tax Platform

From Scratch

Ten product challenges for a client and CPA tax platform that doesn't exist yet. There is no legacy app to inherit and no existing codebase to patch. You're designing and building the first version. Choose the challenges assigned to you, and focus on clarity of thinking, defensible design decisions, and your eye for good user experience.

Overview of Challenges

Each challenge stands alone. Together they cover the core design tensions of a platform built from the ground up: traceability, collaboration, orientation, navigation, roles, status, prioritization, affordances, complexity, and trust in AI.

#	Challenge	In one line
01	Source Document Traceability	Trace every number on the return back to its source
02	Client & CPA Collaboration	Unify fragmented communication around documents and issues
03	Where to Start	A first-time user knows their next action within 10 seconds
04	Getting Lost in the App	Navigation that preserves context across connected objects
05	Role-Aware Experiences	One product, six roles, no confusion
06	Return Status & Progress	Statuses everyone interprets the same way
07	An Actionable Dashboard	Answer “what should I work on right now?”
08	Clickable vs. Editable	A consistent system for interaction affordances
09	Complexity Made Navigable	Deep professional work that stays approachable
10	Trustworthy AI	Transparency that builds confidence without overload

How to Approach This Case Study

This is a greenfield build. Assume nothing exists yet: no legacy screens to patch, no prior design system to match, no existing data model to work around. Every decision, from the first pixel to the last interaction, is yours to make.

THE FRONTEND IS WHAT WE'RE GRADING

Visual design, interaction design, information architecture, and how the product actually feels to use: that's the core of what we're evaluating. A working, clickable prototype tells us far more than a written description or a static mockup ever could.

BEHIND THE FRONTEND, KEEP IT QUICK AND DIRTY

We don't expect production infrastructure, real document parsing, or a working AI model. Hardcoded data, mocked API responses, a small script that fakes the logic: all of that is not just acceptable, it's the ideal way to spend your time. The goal is to prove the concept behind each screen actually works end-to-end, quickly, not to build the real system underneath it.

SIMULATE THE AI

Where a challenge implies AI output, such as extraction, confidence scores, recommendations, or warnings, fabricate plausible fake data for it. We want to see how you'd present AI output and build trust around it, not whether you can build the AI itself.

WHAT GOOD LOOKS LIKE

- A real, working interface, even a rough one, over a set of static images
- Thoughtful, defensible design decisions, explained briefly, over exhaustive polish
- Enough fake data and enough edge cases wired up that the idea is genuinely testable, not just a single happy-path screen
- A short note (a README, a paragraph, whatever's fastest) telling us what's real versus simulated behind the scenes

01

Source Document Traceability

Trace every number on the return back to its source.

THE DESIGN CHALLENGE

On any tax return, a CPA needs to trust every number in front of them. If they can't tell where a figure came from, they either have to take the software's word for it or manually re-derive every value themselves, which defeats the point of automation.

YOUR ASSIGNMENT

Design and build a return review interface, from scratch, that connects:

- The tax return field
- The extracted value shown there
- The source document it came from
- The exact page or section on that document
- Any transformation or calculation applied to get from the source to the field

BEHIND THE SCENES

You don't need real OCR or document parsing. Hardcode a handful of sample documents and fabricate the traceability data linking them to return fields. The interaction model is what we're judging, not extraction accuracy.

WHAT WE ARE EVALUATING

Traceability, transparency, side-by-side review, and how defensible you make AI output feel.

02

Client & CPA Collaboration

Unify fragmented communication around documents and issues.

THE DESIGN CHALLENGE

Tax work involves constant back-and-forth: questions, clarifications, requests for documents, and that conversation easily scatters across email, phone calls, and disconnected tools. Design the collaboration layer for a platform that never has that problem, because it was never built that way.

YOUR ASSIGNMENT

Design and build a communication layer, from scratch, that:

- Connects conversations to specific documents or tax issues
- Distinguishes internal firm notes from client-visible messages
- Tracks outstanding requests
- Shows who owns the next action
- Doesn't just become another generic inbox

BEHIND THE SCENES

Seed a few sample threads and a couple of fake users (one CPA, one client) so the permissions and internal/external distinction are visibly demonstrated. A static set of hardcoded messages with real working UI interactions is enough. No messaging backend required.

WHAT WE ARE EVALUATING

Collaboration design, permissions, contextual communication, and task ownership.

03

Where to Start

A first-time user knows their next action within 10 seconds.

THE DESIGN CHALLENGE

A brand-new client logs into the platform for the very first time. There's no muscle memory, no prior onboarding, nothing to unlearn, just a blank product and a task to complete. Design that first experience.

YOUR ASSIGNMENT

Design and build the first-time experience for a new client, so they understand their next action within 10 seconds, with no training. Decide:

- What they see first
- How navigation is organized from day one
- What's hidden or deferred until it's relevant
- How progress and urgency are communicated
- How the interface changes once onboarding is done

BEHIND THE SCENES

Hardcode a sample “new client” account with a few tasks, documents, and questionnaire items in various states, and build the real onboarding flow around that fake data.

WHAT WE ARE EVALUATING

First-run clarity, information hierarchy, and reducing time-to-first-action.

04

Getting Lost Between Parts of the App

Navigation that preserves context across connected objects.

THE DESIGN CHALLENGE

A client gets a question tied to a specific document. Answering it means moving between messages, the document itself, a questionnaire, and a task list. Design a platform where that movement never costs someone their place.

YOUR ASSIGNMENT

Design and build a navigation model that lets a user move across related documents, questions, tasks, and messages without losing context. Address:

- Global vs. contextual navigation
- Breadcrumbs or other orientation tools
- Deep linking
- Returning to a previous workflow
- Showing how different objects connect to each other

BEHIND THE SCENES

Fake the connections between objects (a document linked to a task linked to a message) with hardcoded relationships. The point is demonstrating the navigation pattern working end-to-end, not building a real data model underneath it.

WHAT WE ARE EVALUATING

Navigation design, orientation, and preserving context across a connected workflow.

05

Role-Aware Experiences

One product, six roles, no confusion.

THE DESIGN CHALLENGE

The same platform will serve individual taxpayers, business owners, tax preparers, reviewers, firm administrators, and seasonal staff. Design it so each of them gets an experience that feels built for them, without splintering into six separate products.

YOUR ASSIGNMENT

Design and build a role-aware frontend architecture. Show:

- How navigation changes by role
- How permissions are communicated
- How a user with multiple roles switches context
- How the system stays one cohesive product across six roles
- How it handles a firm employee who also has a personal return in the system

BEHIND THE SCENES

Hardcode two or three sample roles or logins you can switch between (a preparer view and a client view is a good minimum) to demonstrate how the same shell adapts. No real auth or permissions system required.

WHAT WE ARE EVALUATING

Role architecture, permission clarity, and context-switching design.

06

Return Status & Progress

Statuses everyone interprets the same way.

THE DESIGN CHALLENGE

“In Progress.” “Pending Review.” “Open Items.” Status labels like these mean different things to different people, and clients and staff routinely read them differently. Design status from scratch so it means the same thing to everyone looking at it.

YOUR ASSIGNMENT

Design and build a status and progress experience so clients and firm staff can immediately tell:

- Where the return is in the process
- What's already happened
- What has to happen next
- Who owns the next action
- Whether anything is blocking completion

BEHIND THE SCENES

Hardcode a handful of sample returns at different stages and drive the whole status experience off that fake data. Remember: solve this without exposing unnecessary internal complexity to the client.

WHAT WE ARE EVALUATING

Shared mental models, status legibility, and appropriate detail by audience.

07

An Actionable Dashboard

Answer “what should I work on right now?”

THE DESIGN CHALLENGE

Dashboards are easy to make pretty and easy to make useless. Design the CPA-facing dashboard for a brand-new platform so staff actually use it instead of falling back to a spreadsheet.

YOUR ASSIGNMENT

Design and build a dashboard organized around decisions and actions, not reporting. Determine:

- What belongs on the landing page
- How the most urgent work surfaces
- How someone moves from a summary into action
- How it supports both managers and individual preparers
- How it stays usable when someone owns hundreds of returns

BEHIND THE SCENES

Generate a batch of fake returns and tasks with varied urgency and status, and build real prioritization logic against that mock dataset. A simple script that ranks and filters the fake data is enough.

WHAT WE ARE EVALUATING

Action-orientation, prioritization logic, and dashboard information design.

08

Clickable vs. Editable

A consistent system for interaction affordances.

THE DESIGN CHALLENGE

A platform like this will show AI-generated values, extracted data, calculations, reviewer comments, and client answers side by side, some editable, some requiring approval, some permanently read-only. Design the visual language for that from the very first screen you build.

YOUR ASSIGNMENT

Design and build a consistent interaction system that makes immediately clear:

- What can be clicked
- What can be edited
- What is AI-generated
- What has been verified
- What requires approval
- What can't be changed, and why

BEHIND THE SCENES

Fake the different data states (AI-generated, verified, locked, editable) with hardcoded sample fields so the visual system has real variety to prove itself against. Demonstrate it across several different screens, not just one component.

WHAT WE ARE EVALUATING

Affordance clarity and consistency of the interaction system across contexts.

09

Complexity Made Navigable

Deep professional work that stays approachable.

THE DESIGN CHALLENGE

A single return can involve hundreds of documents, questions, calculations, warnings, and messages. Design a platform that can hold all of that without either overwhelming people or hiding what they need.

YOUR ASSIGNMENT

Design and build a frontend structure that supports deep professional work while staying approachable for occasional users. Demonstrate:

- Progressive disclosure
- Search and filtering
- Hierarchy of information
- Summary vs. detail views
- Persistent context
- Moving between high-level review and source-level detail

BEHIND THE SCENES

Generate a large fake dataset (hundreds of mock documents or items) so your search, filtering, and hierarchy get tested against real volume, not just a handful of demo rows. The challenge isn't reducing the underlying complexity. It's making it navigable.

WHAT WE ARE EVALUATING

Progressive disclosure, information hierarchy, and handling of scale.

10

Trustworthy AI

Transparency that builds confidence without overload.

THE DESIGN CHALLENGE

This platform will lean on AI for recommendations, confidence scores, warnings, extracted values, and suggested corrections. Design the interaction model for that AI from scratch, so people trust it enough to actually use it instead of ignoring it or re-checking everything by hand.

YOUR ASSIGNMENT

Design and build an AI interaction model that helps users understand:

- What the AI did
- Why it made a recommendation
- What evidence supports it
- What uncertainty exists
- What action they should take
- How to correct the AI without breaking their workflow

BEHIND THE SCENES

Fake the AI outputs (recommendations, confidence scores, corrections) with a mocked response format. A stub function returning plausible fake JSON is enough to power a genuinely working UI. Balance transparency with simplicity: showing every technical detail is not an acceptable solution.

WHAT WE ARE EVALUATING

Trust design, appropriate transparency, and correction workflows.

What to Submit

A working prototype (hosted link or a repo we can run locally) covering the challenges assigned to you. It doesn't need to be complete or production-ready. It needs to be real enough to click through.

A short note (README or a few sentences) telling us what's genuinely wired up versus simulated behind the scenes, and any decisions you'd want to explain.